

NETHMI WIJEWEERA

Undergraduate, University of Moratuwa, Faculty of IT

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PROFESSIONAL SUMMARY

Dedicated IT undergraduate and aspiring Software Engineer with a strong foundation in Artificial Intelligence and Full-Stack Development. Proven ability to build scalable backend services, optimize data management, and develop machine learning models through industry internship and academic project experience. Collaborative team player with strong adaptability and a problem-solving mindset, effectively applying technical skills in dynamic environments.

EDUCATION

University of Moratuwa - Faculty of Information Technology	2022 - Present
<i>B.Sc. (Hons) Information Technology</i>	
• Current CGPA: 3.50/4.00 • Dean's List: Semester 3 (3.94 GPA), Semester 4 (3.84 GPA)	
Sirimavo Bandaranaike Vidyalaya - Colombo 07	2011 - 2019
<i>G.C.E. Advanced Level - Physical Science (2020) Z-Score: 1.5252</i>	
<i>G.C.E. Ordinary Level (2016) 9 A's</i>	

WORK EXPERIENCE

Agrithmics (Pvt) Ltd , Sri Lanka	Feb 2025 - Aug 2025
<i>Full-Stack Developer Intern (Onsite)</i>	
• Designed and developed scalable front-end components using React and backend services in ASP.NET Core following Clean Architecture principles. • Implemented efficient data management using MS SQL Server and optimized performance with Redis caching. • Integrated RabbitMQ to enable asynchronous messaging and event-driven workflows across services. • Worked in an Agile development environment, participating in sprint planning, task estimation, and iterative feature delivery. • Collaborated closely with team members using Azure Repos, contributing to version control, code quality, and deployment readiness through peer reviews.	

TECHNICAL SKILLS

Programming Languages: C, Java, JavaScript, C#, Python
Frontend Technologies: HTML, CSS, React.js, Bootstrap, Material-UI
Backend and APIs: ASP.NET Core Web API, RESTful API Development, Dapper ORM, FastAPI, Spring Boot
Databases: MS SQL Server, MySQL, Redis
Tools and QA: Git, GitHub, Azure DevOps, RabbitMQ, Selenium, RestAssured, Cucumber

PROJECTS

Computer Vision-Based Identification of Ingredients in Ayurvedic Powder Mixtures	Dec 2025 - Present
Final Year Project	
• Developed an image-based system to identify ingredients in Ayurvedic powder mixtures. • Implemented particle size-based categorization (Fine-2 and Fine-3) to improve analysis accuracy. • Applied feature extraction methods including shape, surface texture, and color pattern analysis. • Built and trained machine learning models for multi-label ingredient classification. • Tech Stack: Python, OpenCV, TensorFlow	
AgriGen ERP System Internship Project	Feb 2025 - Aug 2025
• Developed an enterprise resource planning system as part of internship work, focusing on scalable backend services and modular front-end architecture. • Utilized React, .NET, MS SQL Server, Redis, RabbitMQ, and Clean Architecture principles. • Tech Stack: React.js, ASP.NET Core, Dapper, C#, RabbitMQ, Azure DevOps, MS SQL Server, Material-UI	

Healthcare Center Management System Second Year Group Project	Sep 2023 - Aug 2024
- Developed a healthcare center management system to enhance overall operational efficiency. - Implemented modules for patient registration, billing, pharmacy management, and appointment scheduling. - Designed and developed a comprehensive doctor interface with role-based functionalities. - Enhanced system usability and efficiency by aligning features with practical operational needs. Tech Stack: <i>React, ASP.NET, Material-UI, MS SQL</i>	
Full-Stack E-Commerce Platform Individual Project	Jan 2026 - Feb 2026
- Implemented JWT-based user authentication with registration, login, and role-based access (USER / ADMIN). - Built secure routes and an admin dashboard with a Spring Boot REST backend for product and inventory management. - Developed order processing, order history tracking, and admin fulfillment management features. Tech Stack: <i>Spring Boot, React, JWT, Axios, Material-UI, MySQL</i>	
LLM-Based Learning Assistant for A/L Biology Students Individual Project	Nov 2025 - Dec 2025
- An LLM-based chat application to answer questions related to the G.C.E A/L Biology syllabus. - This application provides answers to user questions based on the resource books published by the National Institute of Education. Tech Stack: <i>Python, LangChain, Hugging Face, LLMs, FAISS, FastAPI, React</i>	
Used Motorcycle Price Predictor Individual Project	Jan 2025 - Feb 2025
- Developed a system to scrape 1,300+ listings from ikman.lk and predict motorcycle prices. - Integrated SHAP to provide transparent, feature-level justifications for every prediction, ensuring model interpretability. - Built a Streamlit web application allowing users to receive real-time price estimates based on vehicle specifications. Tech Stack: <i>Python, XGBoost, Scikit-learn, SHAP, Streamlit, BeautifulSoup, Pandas</i>	
Movie Sentiment Analysis System Individual Project	Feb 2025 - Aug 2025
- Built a system to classify movie review sentiments (positive/negative) using NLP techniques. - Developed machine learning models using SVM and Logistic Regression with high accuracy. - Processed and cleaned text data, extracted features using TF-IDF, and balanced dataset with SMOTE. Tech Stack: <i>Python, Scikit-learn, NLTK, SMOTE, NumPy, Pandas, Matplotlib</i>	
Smart First Aid Advisor Individual Project	Oct 2025 - Nov 2025
- Developed intelligent first-aid advisor using a rule-based expert system with 20+ emergency scenarios. - Implemented dynamic question-based diagnosis with conditional logic for customized guidance. - Designed intuitive GUI using Tkinter with session logging and emergency care recommendations. Tech Stack: <i>Python, Tkinter, Expert Systems, Knowledge-Based AI</i>	
Automated Fruit Juice Machine First Year Hardware Project	Aug 2022 - Aug 2023
- Engineered automated dispensing system with customizable fruit selection and adjustable sugar levels. - Programmed Arduino microcontroller for automated operation with integrated sensors and actuators. Tech Stack: <i>Arduino, C</i>	

CERTIFICATIONS

Diploma in DevOps Engineering (Kubernetes, Docker, Google Cloud) - Alison
Machine Learning Specialization - Coursera
Object-Oriented Programming with Java - Coursera
Web Development for Beginners - Open University of Moratuwa

REFERENCES

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